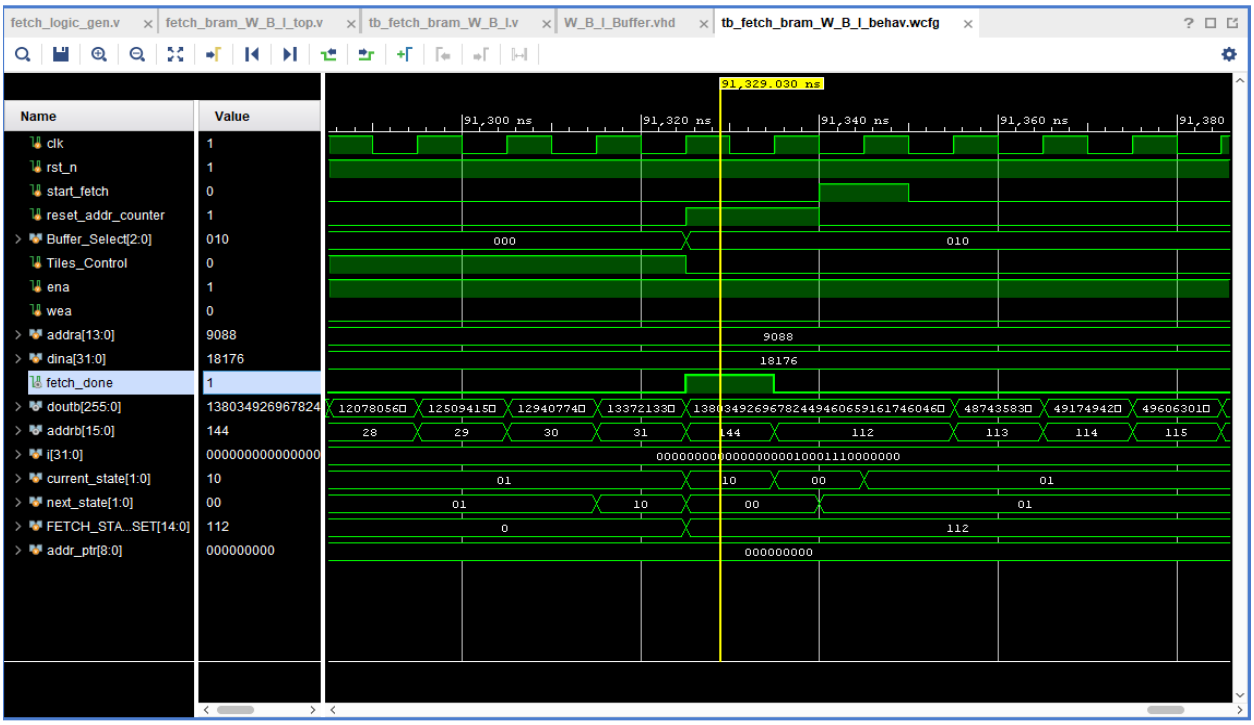
Fetch logic is responsible for generating the address to the BRAMs to get tiles and pass it to the SA,

Treating it as a black box here is a table for its interface:

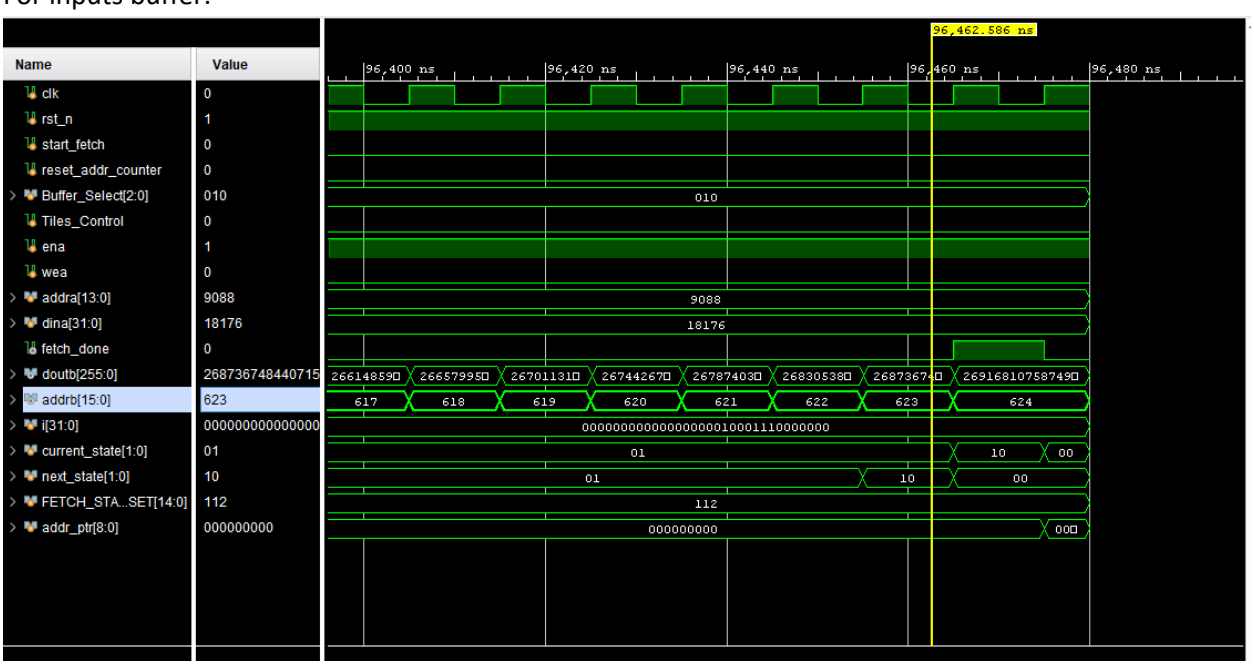
Direction	signal	Description
input	Clk	system clock
input	Reset	System asynchronous active low reset
input	Start_fetch	takes a start pulse and begins fetching from the BRAM we choose
input	Reset_addr_counter	reset_counter input to reset the address pointer
input	Double_buffer	Signal from main control to start address from double buffering addresses
input	Buffer_Select	buffer select input to select which buffer we will fetch from (W, B, I, Q, K, V)
input	Tiles_Control	tiles_control input to choose whether we will fetch 32 (W, B) rows per tile or 512 rows per tile (I, Q, K, V)
output	Bram_addr	It outputs the address we will read from to the chosen BRAM
output	Bram_en	Outputs enable to BRAM (vivado restriction)
output	Fetch_done	Outputs fetch_done signal (pulse) to SA indicating the tile is done
output	busy	Outputs busy signal indicating we are fetching

For weights buffer:



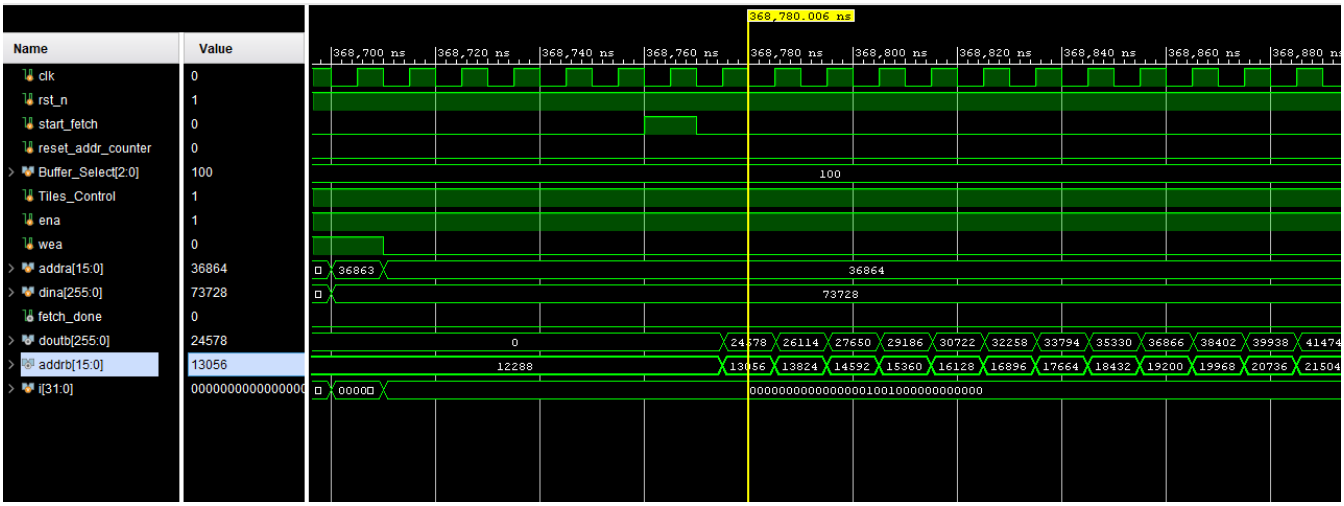
- Here we see that the valid pulse arises one cycle after finishing and we need another cycle to assert the start signal for the next tile (input tile)
- Notice how the address for a weight tile ends at address 31 as we are only using 32 rows tile fetch
- Notice how the starting address for the input tile (I) begins at 112

For inputs buffer:

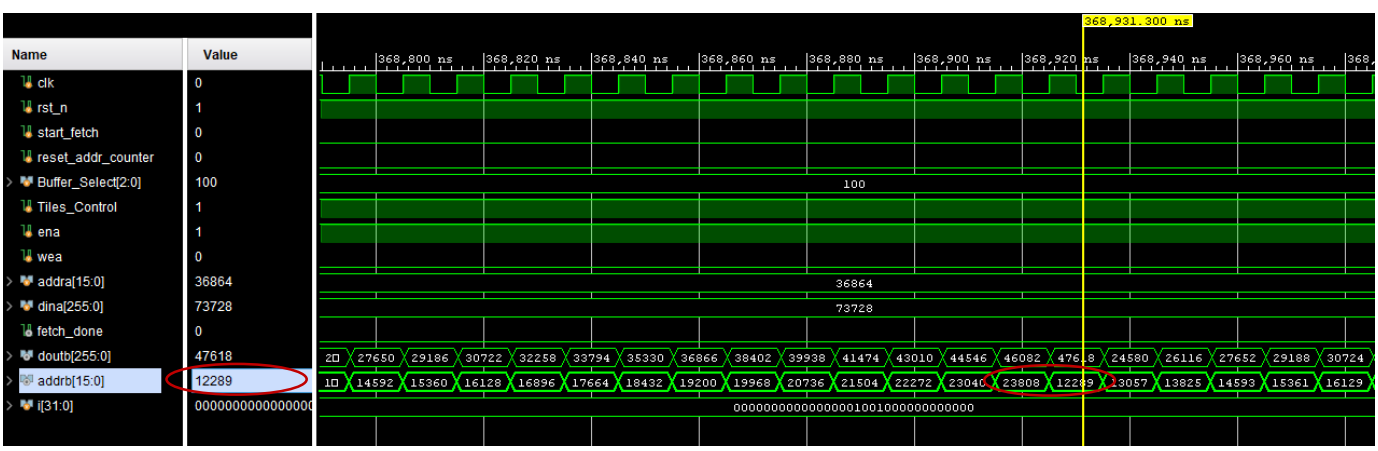


- For input tile (512) notice how we ended our fetching at address 623 as (512 +112) = 624 but we include the zero address, but what is 112? We will have a detailed diagram for the buffers with their values, but for now this is the starting address for the Input buffer

For K buffer:



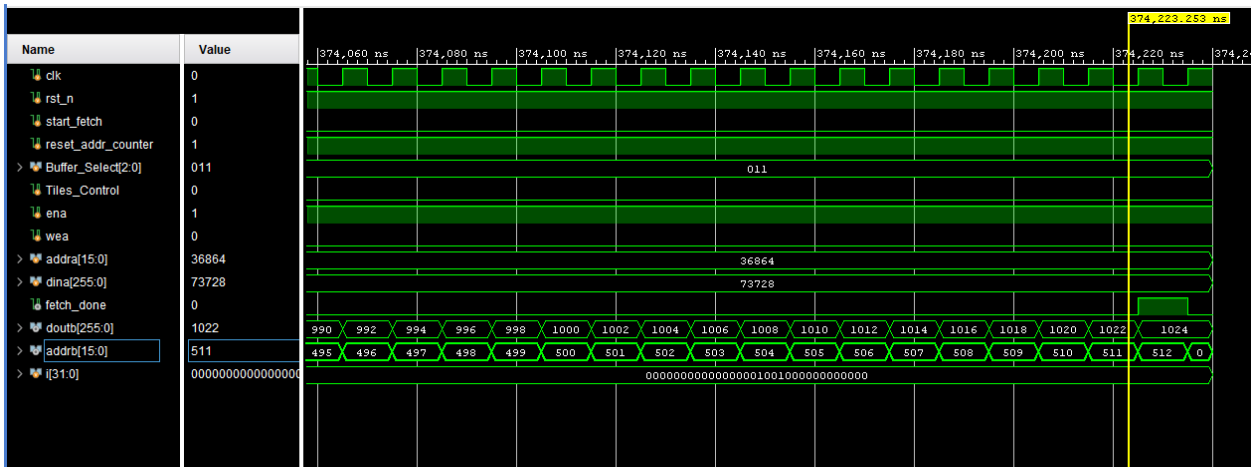
- In multi head attention equation we need the K buffer specifically to get fetched in a transposition manner
- Notice that the first address is 12288 then the next address is 13056 why? As if we do 12288 – 13056 = 768, and this way we get the elements transposed from K buffer considering our K buffer is 512 x 768 matrix
- So we are incrementing the address by 768 to take the elements in the first column then moving to the second and so on
- If we calculate the no. of the elements we will take from the first column it will be: 12288 (buffer size) / 768 = 16 so we will need to pass the first 2 columns elements



- Here is the point we're talking about, when we reached the 23808 address if we increment for another 768 we will end up in 24576 and that's the first address of the V buffer so we are no longer in our K buffer anymore

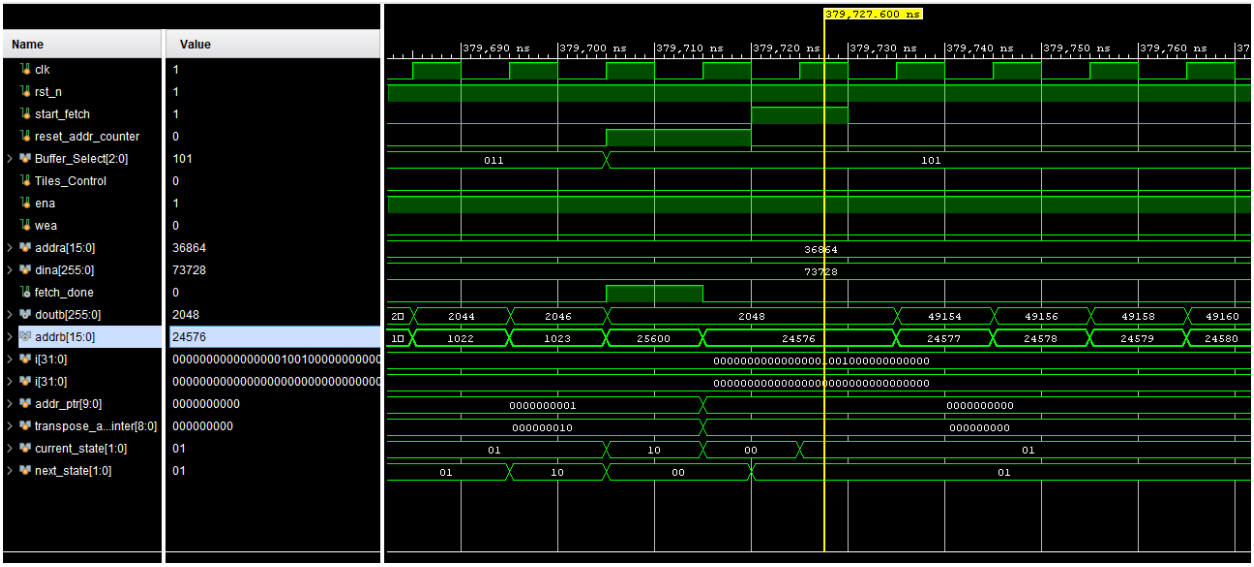
- So when we reach the 23808 we go up again to the first element for the second column which is at the address 12289 and we increment 768 all along to reach to the bottom of this column and here are the 32 stationary elements for our Systolic array

For Q buffer:



- We've started at address 0 and we took a 512 element tile like normal

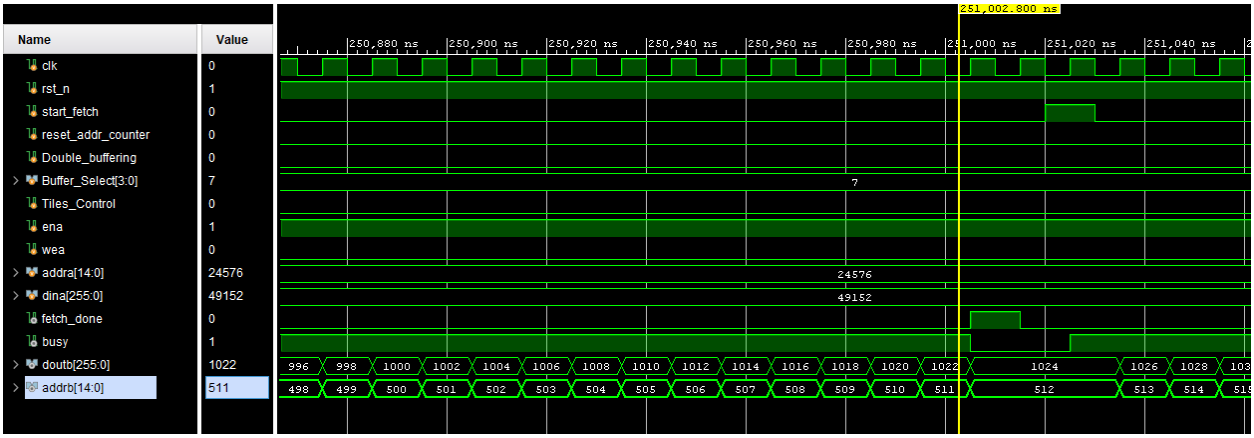
For V buffer:



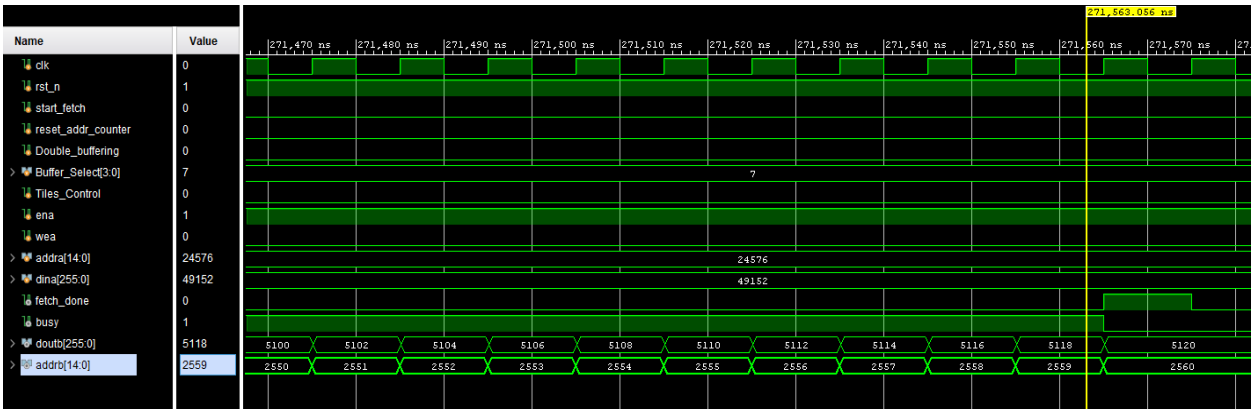
- We begin at 24576 address and we tile a normal 512 element tile

For Kt x Q buffer:

For S x V matrix:

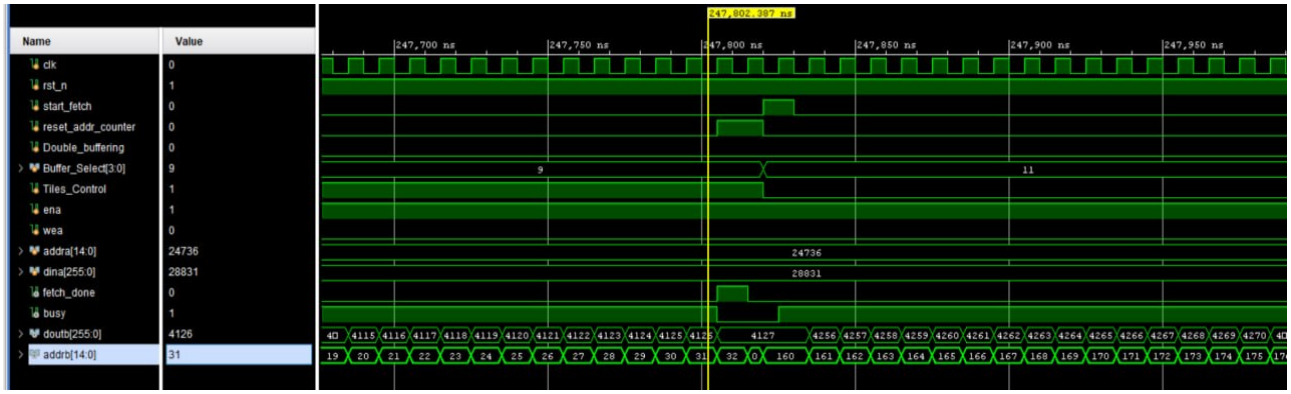


- For the output matrix from the softmax (matrix S) and the V we will tile at 512 clock cycles as we will multiply the S x V matrix by another weights to get the final output matrix of our multi-head attention

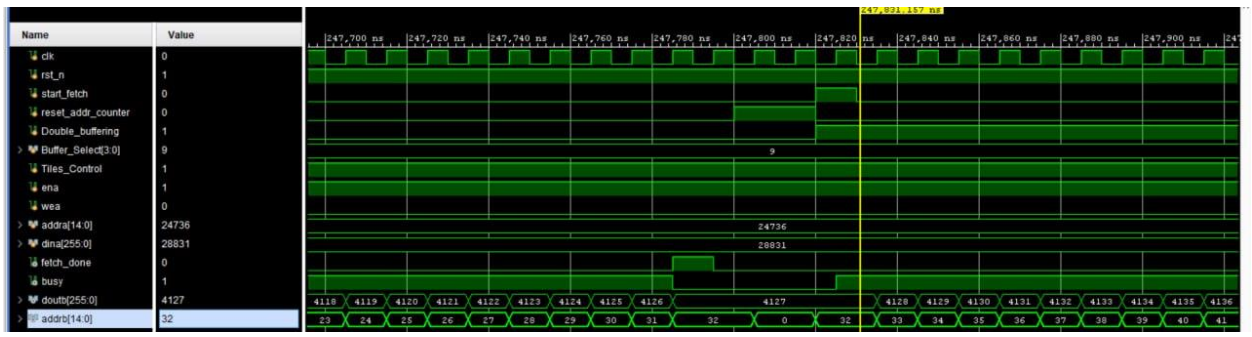


- We did fetch 5 times and every fetch we are reading 512 x 32 elements in 512 cycles, so our address when fifth time will be 5 x 512 – 1 = 2559

For FFN Weights buffer:

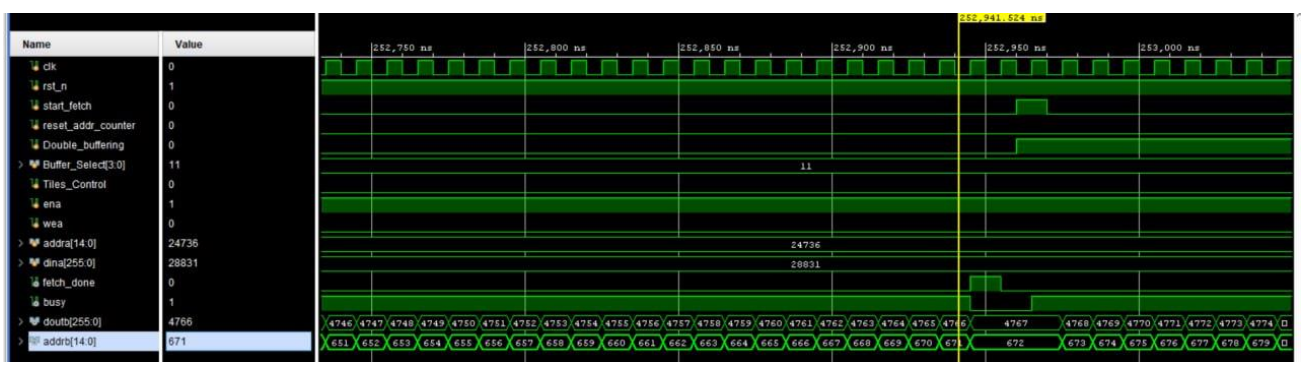


- It's just the same as the multi-head attention weights with the same double buffering logic so we tile at 32 cycle

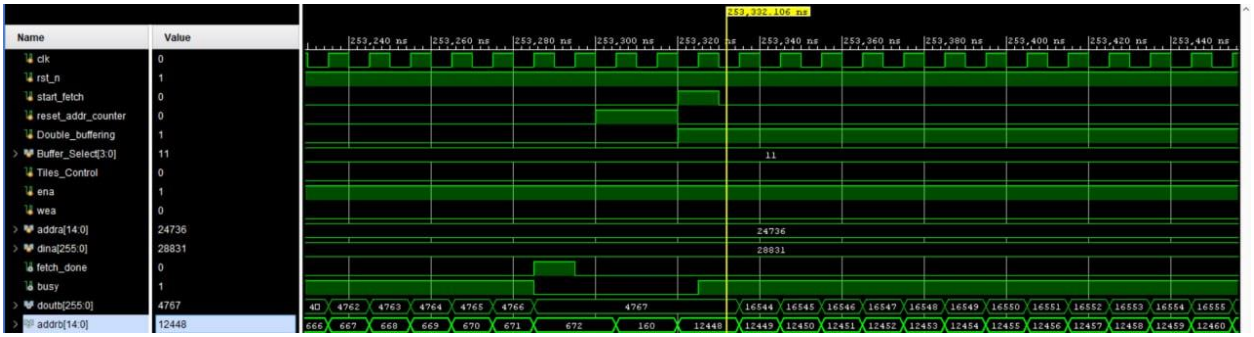


- For double buffering we begin fetching at 32

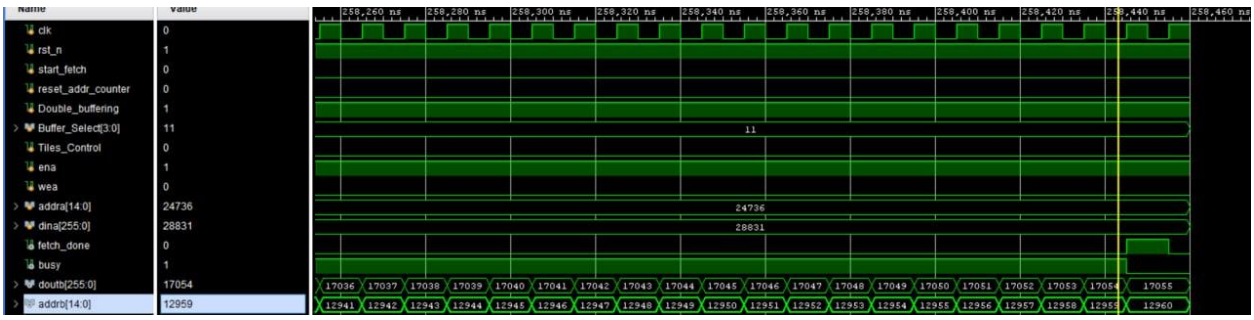
For FFN Input buffer:



- For our input buffer in feed forward network it's also the same as our multi head input buffer, we are tiling 512 so 512 + 160 = 672, as 160 is the start offset for our input buffer

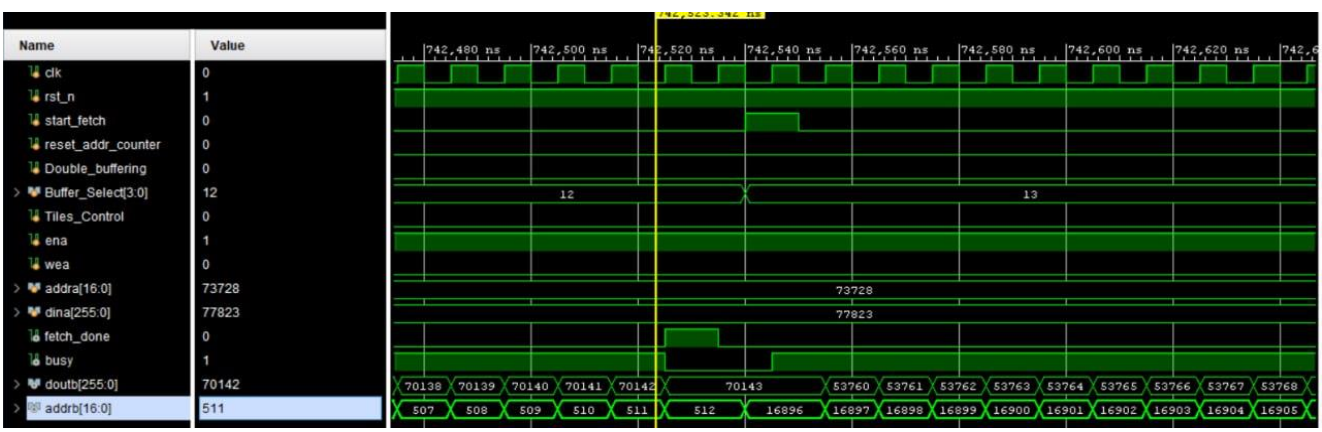


- For input double buffering the first address will be 12488 as 160 + 12288 = 12488 and we will tile 512



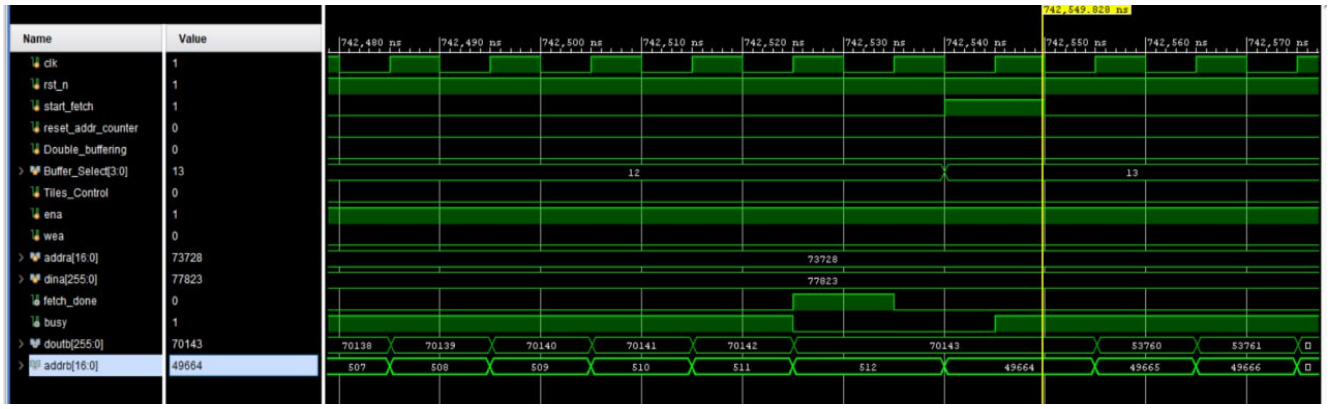
- After tiling 512 so the final address will be 12959

For FFN intermediate buffer:



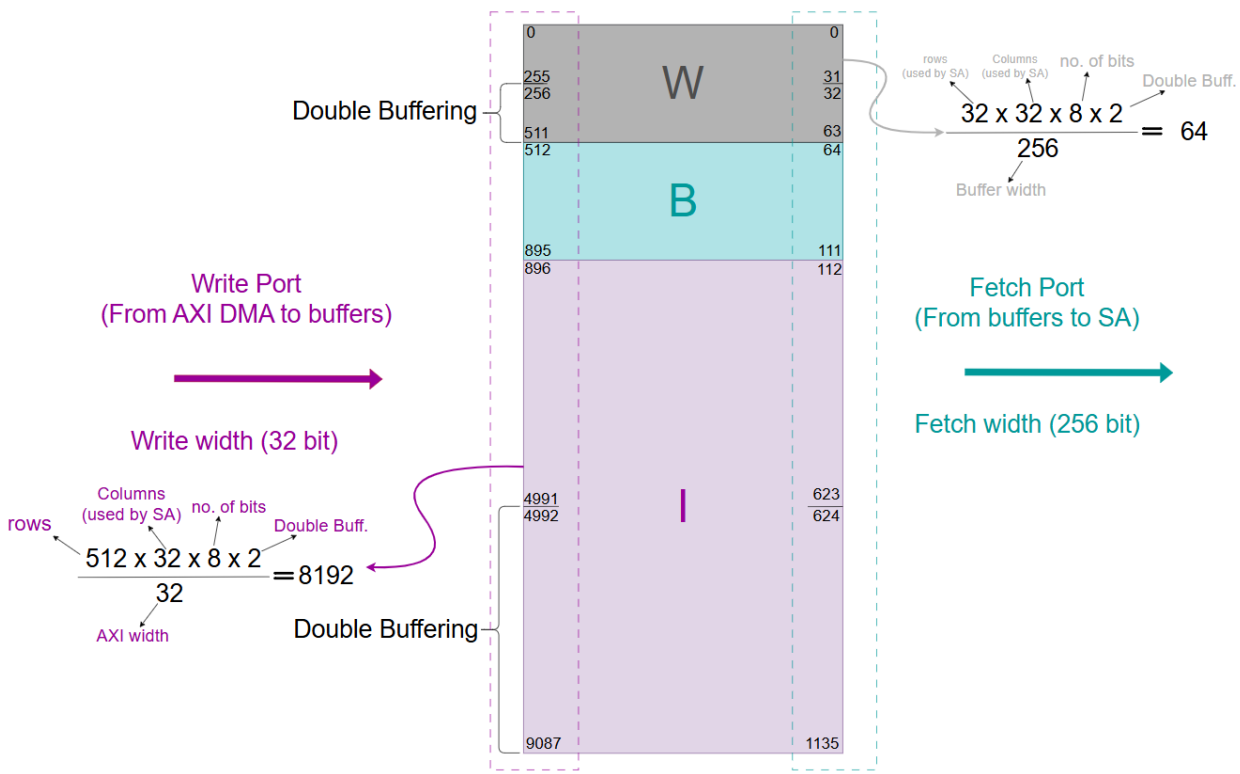
- Same as many we tile 512

For output buffer:



- It will begin at offset 49152 as our intermediate buffer is 512 x 3072 x 8 / 256 = 49152, so we will begin at this address and then tile 512

W B I Buffer:



In general, our weights is (512, 512) but we made the buffer only 32 as our systolic array is weight stationary type and it's size is (32, 32)

Component Name W_B_I_Buffer

Basic | Port A Options | Port B Options | Other Options | **Summary**

Information

Memory Type: Simple Dual Port RAM
Block RAM resource(s) (18K BRAMs): 4
Block RAM resource(s) (36K BRAMs): 9
Total Port B Read Latency (From Rising Edge of Read Clock): 1 Clock Cycle(s)
Address Width A: 14
Address Width B : 11

Component Name Q_K_V_buffer

Basic | Port A Options | Port B Options | Other Options | **Summary**

Information

Memory Type: Simple Dual Port RAM
Block RAM resource(s) (18K BRAMs): 9
Block RAM resource(s) (36K BRAMs): 252
Total Port B Read Latency (From Rising Edge of Read Clock): 1 Clock Cycle(s)
Address Width A: 16
Address Width B : 16

Assumptions:

- The writing AXI bus width is 32 bit
- the output width bus is 256 used by the write logic, so we will give 32 elements to the SA in one cycle